

Line-by-Line Mathematical Guide for HAETAE_Params.ec

Generated HAETAE EasyCrypt proof guide

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File Role

Parameter modes and constants.

How To Read This Script

Read this file as the shared numerical vocabulary used by algebra, distributions, rejection sampling, and reductions.

Line-by-Line Mathematical Reading

Each row preserves one EasyCrypt source line and gives the intended mathematical or proof-script reading. Blank rows are kept because they mark proof structure in long EasyCrypt developments.

Line	EasyCrypt source	Mathematical reading
1	<code>require import AllCore.</code>	Imports AllCore. Mathematically, this exposes earlier carrier sets, operations, games, and lemmas that the current file may cite.
2	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
3	<code>theory HAETAE_Params.</code>	Begins the namespace HAETAE_Params, grouping the declarations as one checked theory.
4	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
5	<code>op seedbytes : int = 32.</code>	Defines integer-valued parameter seedbytes, used for dimensions, budgets, support sizes, or bounds.
6	<code>op crhbytes : int = 64.</code>	Defines integer-valued parameter crhbytes, used for dimensions, budgets, support sizes, or bounds.
7	<code>op n : int = 256.</code>	Defines integer-valued parameter n, used for dimensions, budgets, support sizes, or bounds.
8	<code>op q : int = 64513.</code>	Defines integer-valued parameter q, used for dimensions, budgets, support sizes, or bounds.
9	<code>op dq : int = 2 * q.</code>	Defines integer-valued parameter dq, used for dimensions, budgets, support sizes, or bounds.
10	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
11	<code>type mode = [Mode2 Mode3 Mode5].</code>	Declares carrier set mode. EasyCrypt will type-check all later functions and games over this mathematical domain.
12	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
13	<code>op mode_k (md : mode) : int =</code>	Defines integer-valued parameter mode_k, used for dimensions, budgets, support sizes, or bounds.
14	<code>with md = Mode2 => 2</code>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.

Line	EasyCrypt source	Mathematical reading
15	with md = Mode3 => 3	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
16	with md = Mode5 => 4.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
17	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
18	op mode_l (md : mode) : int =	Defines integer-valued parameter mode_l, used for dimensions, budgets, support sizes, or bounds.
19	with md = Mode2 => 4	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
20	with md = Mode3 => 6	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
21	with md = Mode5 => 7.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
22	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
23	op mode_tau (md : mode) : int =	Defines integer-valued parameter mode_tau, used for dimensions, budgets, support sizes, or bounds.
24	with md = Mode2 => 58	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
25	with md = Mode3 => 80	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
26	with md = Mode5 => 128.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
27	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
28	op mode_b0sq (md : mode) : int =	Defines integer-valued parameter mode_b0sq, used for dimensions, budgets, support sizes, or bounds.
29	with md = Mode2 => 96944109	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
30	with md = Mode3 => 335438492	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
31	with md = Mode5 => 499239142.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
32	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
33	op mode_b1sq (md : mode) : int =	Defines integer-valued parameter mode_b1sq, used for dimensions, budgets, support sizes, or bounds.
34	with md = Mode2 => 96805527	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
35	with md = Mode3 => 335171879	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
36	with md = Mode5 => 498849991.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
37	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
38	op mode_b2sq (md : mode) : int =	Defines integer-valued parameter mode_b2sq, used for dimensions, budgets, support sizes, or bounds.
39	with md = Mode2 => 163265017	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
40	with md = Mode3 => 479901314	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
41	with md = Mode5 => 597386433.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
42	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
43	op mode_ln (_ : mode) : int = 8192.	Defines integer-valued parameter mode_ln, used for dimensions, budgets, support sizes, or bounds.
44	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
45	op mode_alpha_hint (md : mode) : int =	Defines integer-valued parameter mode_alpha_hint, used for dimensions, budgets, support sizes, or bounds.
46	with md = Mode2 => 512	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
47	with md = Mode3 => 512	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.

Line	EasyCrypt source	Mathematical reading
48	with md = Mode5 => 256.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
49	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
50	op mode_m (md : mode) : int = mode_l md - 1.	Defines integer-valued parameter mode_m, used for dimensions, budgets, support sizes, or bounds.
51	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
52	op mode_crypto_bytes (md : mode) : int =	Defines integer-valued parameter mode_crypto_bytes, used for dimensions, budgets, support sizes, or bounds.
53	with md = Mode2 => 1474	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
54	with md = Mode3 => 2349	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
55	with md = Mode5 => 2948.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
56	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
57	op mode_polyq_packedbytes (md : mode) : int =	Defines integer-valued parameter mode_polyq_packedbytes, used for dimensions, budgets, support sizes, or bounds.
58	with md = Mode2 => 480	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
59	with md = Mode3 => 480	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
60	with md = Mode5 => 512.	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
61	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
62	op mode_publickeybytes (md : mode) : int =	Defines integer-valued parameter mode_publickeybytes, used for dimensions, budgets, support sizes, or bounds.
63	seedbytes + mode_k md * mode_polyq_packedbytes md.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
64	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
65	lemma seedbytes_gt0 : 0 < seedbytes by rewrite /seedbytes.	States checked lemma seedbytes_gt0. The following proof script must derive this proposition from prior definitions and lemmas.
66	lemma crhbytes_gt0 : 0 < crhbytes by rewrite /crhbytes.	States checked lemma crhbytes_gt0. The following proof script must derive this proposition from prior definitions and lemmas.
67	lemma n_gt0 : 0 < n by rewrite /n.	States checked lemma n_gt0. The following proof script must derive this proposition from prior definitions and lemmas.
68	lemma q_gt0 : 0 < q by rewrite /q.	States checked lemma q_gt0. The following proof script must derive this proposition from prior definitions and lemmas.
69	lemma dqE : dq = 129026 by rewrite /dq /q.	States checked lemma dqE. The following proof script must derive this proposition from prior definitions and lemmas.
70	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
71	lemma mode_k_gt0 md : 0 < mode_k md.	States checked lemma mode_k_gt0. The following proof script must derive this proposition from prior definitions and lemmas.
72	proof. by case md. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
73	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
74	lemma mode_l_gt0 md : 0 < mode_l md.	States checked lemma mode_l_gt0. The following proof script must derive this proposition from prior definitions and lemmas.
75	proof. by case md. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
76	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
77	lemma mode_tau_gt0 md : 0 < mode_tau md.	States checked lemma mode_tau_gt0. The following proof script must derive this proposition from prior definitions and lemmas.
78	proof. by case md. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
79	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
80	lemma mode_tau_le_n md : mode_tau md <= n.	States checked lemma mode_tau_le_n. The following proof script must derive this proposition from prior definitions and lemmas.

Line	EasyCrypt source	Mathematical reading
81	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
82	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
83	<code>lemma mode_b0sq_gt0 md : 0 < mode_b0sq md.</code>	States checked lemma <code>mode_b0sq_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
84	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
85	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
86	<code>lemma mode_b1sq_gt0 md : 0 < mode_b1sq md.</code>	States checked lemma <code>mode_b1sq_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
87	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
88	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
89	<code>lemma mode_b2sq_gt0 md : 0 < mode_b2sq md.</code>	States checked lemma <code>mode_b2sq_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
90	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
91	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
92	<code>lemma mode_ln_gt0 md : 0 < mode_ln md.</code>	States checked lemma <code>mode_ln_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
93	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
94	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
95	<code>lemma mode_alpha_hint_gt0 md : 0 < mode_alpha_hint md.</code>	States checked lemma <code>mode_alpha_hint_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
96	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
97	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
98	<code>lemma mode_m_gt0 md : 0 < mode_m md.</code>	States checked lemma <code>mode_m_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
99	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
100	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
101	<code>lemma mode_crypto_bytes_gt0 md : 0 < mode_crypto_bytes md.</code>	States checked lemma <code>mode_crypto_bytes_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
102	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
103	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
104	<code>lemma mode_polyq_packedbytes_gt0 md : 0 < mode_polyq_packedbytes md.</code>	States checked lemma <code>mode_polyq_packedbytes_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
105	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
106	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
107	<code>lemma n_le_mode_polyq_packedbytes md : n <= mode_polyq_packedbytes md.</code>	States checked lemma <code>n_le_mode_polyq_packedbytes</code> . The following proof script must derive this proposition from prior definitions and lemmas.
108	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
109	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
110	<code>lemma mode_publickeybytes_gt0 md : 0 < mode_publickeybytes md.</code>	States checked lemma <code>mode_publickeybytes_gt0</code> . The following proof script must derive this proposition from prior definitions and lemmas.
111	<code>proof. by case md. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
112	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
113	<code>end HAETAET_Params.</code>	Closes the current theory or module block.